Employee Performance Mapping

Course-End Project 1 | SQL

ScienceQtech is a data science startup and the HR analysis team needs reports on employee details, performance and projects for the annual appraisal cycle. This document has the SQL queries written for each requirement in the problem statement, along with screenshots of the output and a short note on what the result shows. All queries were run on MySQL 8.0 against the employee database.

Q1. Creating the database and importing the data

Task: Create a database named employee, then import data_science_team.csv, proj_table.csv, and emp_record_table.csv into the employee database.

Query:

```
CREATE DATABASE employee;
USE employee;
```

After creating the database, I imported data_science_team.csv, proj_table.csv and emp_record_table.csv into it and checked that all three tables were created.

Query:

```
SHOW TABLES;
```

Output:

Tables_in_employee
data_science_team
emp_record_table
proj_table

Query:

```
SELECT * FROM emp_record_table;
```

Output:

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT	SALARY	EMP_RATING	MANAGER_ID	PROJ_ID
E001	Arthur	Black	M	PRESIDENT	ALL	20	USA	NORTH AMERICA	16500	5	NULL	NULL
E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA	8500	3	E103	P105
E010	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE	9000	2	E428	P204
E052	Dianna	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	6	CANADA	NORTH AMERICA	5500	5	E083	P103
E057	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA	7700	1	E083	P302
E083	Patrick	Voltz	M	MANAGER	HEALTHCARE	15	USA	NORTH AMERICA	9500	5	E001	NULL
E103	Emily	Grove	F	MANAGER	FINANCE	14	CANADA	NORTH AMERICA	10500	4	E001	NULL
E204	Karene	Nowak	F	SENIOR DATA SCIENTIST	AUTOMOTIVE	8	GERMANY	EUROPE	7500	5	E428	P204
E245	Nian	Zhen	M	SENIOR DATA SCIENTIST	RETAIL	6	CHINA	ASIA	6500	2	E583	P109
E260	Roy	Collins	M	SENIOR DATA SCIENTIST	RETAIL	7	INDIA	ASIA	7000	3	E583	NA
E403	Steve	Hoffman	M	ASSOCIATE DATA SCIENTIST	FINANCE	4	USA	NORTH AMERICA	5000	3	E103	P105
E428	Pete	Allen	M	MANAGER	AUTOMOTIVE	14	GERMANY	EUROPE	11000	4	E001	NULL
E478	David	Smith	M	ASSOCIATE DATA SCIENTIST	RETAIL	3	COLOMBIA	SOUTH AMERICA	4000	4	E583	P109
E505	Chad	Wilson	M	ASSOCIATE DATA SCIENTIST	HEALTHCARE	5	CANADA	NORTH AMERICA	5000	2	E083	P103
E532	Claire	Brennan	F	ASSOCIATE DATA SCIENTIST	AUTOMOTIVE	3	GERMANY	EUROPE	4300	1	E428	P204
E583	Janet	Hale	F	MANAGER	RETAIL	14	COLOMBIA	SOUTH AMERICA	10000	2	E001	NULL
E612	Tracy	Norris	F	MANAGER	RETAIL	13	INDIA	ASIA	8500	4	E001	NULL
E620	Katrina	Allen	F	JUNIOR DATA SCIENTIST	RETAIL	2	INDIA	ASIA	3000	1	E612	P406
E640	Jenifer	Jhones	F	JUNIOR DATA SCIENTIST	RETAIL	1	COLOMBIA	SOUTH AMERICA	2800	4	E612	P406

Query:

```
SELECT * FROM proj_table;
```

Output:

PROJECT_ID	PROJ_NAME	DOMAIN	START _DATE	CLOSURE_DATE	DEV_QTR	STATUS
P103	Drug Discovery	HEALTHCARE	04-06-2021	6/20/2021	Q1	DONE
P105	Fraud Detection	FINANCE	04-11-2021	6/25/2021	Q1	DONE
P109	Market Basket Analysis	RETAIL	04-12-2021	6/30/2021	Q1	DELAYED
P204	Supply Chain Management	AUTOMOTIVE	07/15/2021	9/28/2021	Q2	WIP
P302	Early Detection of Lung Cancer	HEALTHCARE	10-08-2021	12/18/2021	Q3	YTS
P406	Customer Sentiment Analysis	RETAIL	07-09-2021	9/24/2021	Q2	WIP

Query:

```
SELECT * FROM data_science_team;
```

Output:

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT
E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA
E010	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE
E052	Dianna	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	6	CANADA	NORTH AMERICA
E057	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA
E204	Karene	Nowak	F	SENIOR DATA SCIENTIST	AUTOMOTIVE	8	GERMANY	EUROPE
E245	Nian	Zhen	M	SENIOR DATA SCIENTIST	RETAIL	6	CHINA	ASIA
E260	Roy	Collins	M	SENIOR DATA SCIENTIST	RETAIL	7	INDIA	ASIA
E403	Steve	Hoffman	M	ASSOCIATE DATA SCIENTIST	FINANCE	4	USA	NORTH AMERICA
E478	David	Smith	M	ASSOCIATE DATA SCIENTIST	RETAIL	3	COLOMBIA	SOUTH AMERICA
E505	Chad	Wilson	M	ASSOCIATE DATA SCIENTIST	HEALTHCARE	5	CANADA	NORTH AMERICA
E532	Claire	Brennan	F	ASSOCIATE DATA SCIENTIST	AUTOMOTIVE	3	GERMANY	EUROPE
E620	Katrina	Allen	F	JUNIOR DATA SCIENTIST	RETAIL	2	INDIA	ASIA
E640	Jenifer	Jhones	F	JUNIOR DATA SCIENTIST	RETAIL	1	COLOMBIA	SOUTH AMERICA

Observation:

All three tables were imported into the employee database. emp_record_table has 19 rows, proj_table has 6 and data_science_team has 13, which matches the csv files. Two small things I noticed in the data: the start date column in proj_table is named START _DATE (there is a space in the csv header), and Roy Collins (E260) has NA as his PROJ_ID while the other employees without a project have NULL.

Q2. Employee list with department details

Task: Fetch EMP_ID, FIRST_NAME, LAST_NAME, GENDER, and DEPARTMENT from the employee record table, and make a list of employees and details of their department.

Query:

```
SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT AS DEPARTMENT
FROM emp_record_table
ORDER BY DEPT, EMP_ID;
```

Output:

	EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPARTMENT
▶	E001	Arthur	Black	M	ALL
	E010	William	Butler	M	AUTOMOTIVE
	E204	Karene	Nowak	F	AUTOMOTIVE
	E428	Pete	Allen	M	AUTOMOTIVE
	E532	Claire	Brennan	F	AUTOMOTIVE
	E005	Eric	Hoffman	M	FINANCE
	E103	Emily	Grove	F	FINANCE
	E403	Steve	Hoffman	M	FINANCE
	E052	Dianna	Wilson	F	HEALTHCARE
	E057	Dorothy	Wilson	F	HEALTHCARE
	E083	Patrick	Voltz	M	HEALTHCARE
	E505	Chad	Wilson	M	HEALTHCARE
	E245	Nian	Zhen	M	RETAIL
	E260	Roy	Collins	M	RETAIL
	E478	David	Smith	M	RETAIL
	E583	Janet	Hale	F	RETAIL
	E612	Tracy	Norris	F	RETAIL
	E620	Katrina	Allen	F	RETAIL
	E640	Jenifer	Jhones	F	RETAIL

Observation:

The list covers all 19 employees. Retail is the largest department with 7 people, Automotive and Healthcare have 4 each and Finance has 3. Arthur Black is the President, so his department shows as ALL.

Q3. Employees by performance rating

Task: Fetch EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPARTMENT, and EMP_RATING if the EMP_RATING is less than two, between two and four, and greater than four.

a) EMP_RATING less than 2

Query:

```
SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT AS DEPARTMENT, EMP_RATING
FROM emp_record_table
WHERE EMP_RATING < 2;
```

Output:

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPARTMENT	EMP_RATING
E057	Dorothy	Wilson	F	HEALTHCARE	1
E532	Claire	Brennan	F	AUTOMOTIVE	1
E620	Katrina	Allen	F	RETAIL	1

b) EMP_RATING between 2 and 4

Query:

```
SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT AS DEPARTMENT, EMP_RATING
FROM emp_record_table
WHERE EMP_RATING BETWEEN 2 AND 4;
```

Output:

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPARTMENT	EMP_RATING
E005	Eric	Hoffman	M	FINANCE	3
E010	William	Butler	M	AUTOMOTIVE	2
E103	Emily	Grove	F	FINANCE	4
E245	Nian	Zhen	M	RETAIL	2
E260	Roy	Collins	M	RETAIL	3
E403	Steve	Hoffman	M	FINANCE	3
E428	Pete	Allen	M	AUTOMOTIVE	4
E478	David	Smith	M	RETAIL	4
E505	Chad	Wilson	M	HEALTHCARE	2
E583	Janet	Hale	F	RETAIL	2
E612	Tracy	Norris	F	RETAIL	4
E640	Jenifer	Jhones	F	RETAIL	4

c) EMP_RATING greater than 4

Query:

```
SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT AS DEPARTMENT, EMP_RATING
FROM emp_record_table
WHERE EMP_RATING > 4;
```

Output:

	EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPARTMENT	EMP_RATING
▶	E001	Arthur	Black	M	ALL	5
	E052	Dianna	Wilson	F	HEALTHCARE	5
	E083	Patrick	Voltz	M	HEALTHCARE	5
	E204	Karene	Nowak	F	AUTOMOTIVE	5

Observation:

3 employees are rated below 2 (all of them have a rating of 1), 12 are between 2 and 4, and 4 are rated above 4 (all of them have a 5). So most of the team is in the middle band. BETWEEN is inclusive, so employees rated exactly 2 or 4 fall in the middle group.

Q4. Employees in leadership positions

Task: Retrieve the employee ID, first name, role, and department of employees who hold leadership positions (Manager, President, or CEO).

Query:

```
SELECT EMP_ID, FIRST_NAME, ROLE, DEPT AS DEPARTMENT
FROM emp_record_table
WHERE ROLE IN ('MANAGER', 'PRESIDENT', 'CEO')
ORDER BY ROLE;
```

Output:

	EMP_ID	FIRST_NAME	ROLE	DEPARTMENT
▶	E083	Patrick	MANAGER	HEALTHCARE
	E103	Emily	MANAGER	FINANCE
	E428	Pete	MANAGER	AUTOMOTIVE
	E583	Janet	MANAGER	RETAIL
	E612	Tracy	MANAGER	RETAIL
	E001	Arthur	PRESIDENT	ALL

Observation:

There are 6 employees in leadership positions: 5 managers and the President, Arthur Black. Nobody in the table has the CEO role, so that part of the condition does not return anything. Retail is the only department with two managers (Janet Hale and Tracy Norris).

Q5. Employee ratings by department

Task: List employee details (EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPARTMENT, EMP_RATING) grouped by department, along with the max emp rating for the department.

Query:

```
SELECT EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPT AS DEPARTMENT, EMP_RATING,
       MAX(EMP_RATING) OVER (PARTITION BY DEPT) AS MAX_DEPT_RATING
FROM emp_record_table
ORDER BY DEPT, EMP_RATING DESC;
```

Output:

	EMP_ID	FIRST_NAME	LAST_NAME	ROLE	DEPARTMENT	EMP_RATING	MAX_DEPT_RATING
▶	E001	Arthur	Black	PRESIDENT	ALL	5	5
	E204	Karene	Nowak	SENIOR DATA SCIENTIST	AUTOMOTIVE	5	5
	E428	Pete	Allen	MANAGER	AUTOMOTIVE	4	5
	E010	William	Butler	LEAD DATA SCIENTIST	AUTOMOTIVE	2	5
	E532	Claire	Brennan	ASSOCIATE DATA SCIENTIST	AUTOMOTIVE	1	5
	E103	Emily	Grove	MANAGER	FINANCE	4	4
	E005	Eric	Hoffman	LEAD DATA SCIENTIST	FINANCE	3	4
	E403	Steve	Hoffman	ASSOCIATE DATA SCIENTIST	FINANCE	3	4
	E052	Dianna	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	5	5
	E083	Patrick	Voltz	MANAGER	HEALTHCARE	5	5
	E505	Chad	Wilson	ASSOCIATE DATA SCIENTIST	HEALTHCARE	2	5
	E057	Dorothy	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	1	5
	E640	Jenifer	Jhones	JUNIOR DATA SCIENTIST	RETAIL	4	4
	E612	Tracy	Norris	MANAGER	RETAIL	4	4
	E478	David	Smith	ASSOCIATE DATA SCIENTIST	RETAIL	4	4
	E260	Roy	Collins	SENIOR DATA SCIENTIST	RETAIL	3	4
	E583	Janet	Hale	MANAGER	RETAIL	2	4
	E245	Nian	Zhen	SENIOR DATA SCIENTIST	RETAIL	2	4
	E620	Katrina	Allen	JUNIOR DATA SCIENTIST	RETAIL	1	4

Observation:

I used MAX() with OVER (PARTITION BY DEPT) instead of a normal GROUP BY, because GROUP BY would reduce the result to one row per department and the employee details would be lost. Automotive and Healthcare both have a top rating of 5, while Finance and Retail top out at 4. Healthcare has the widest spread, from 5 (Dianna Wilson, Patrick Voltz) down to 1 (Dorothy Wilson).

Q6. Minimum and maximum salary by role

Task: Find the minimum and maximum salary for employees grouped by their role.

Query:

```
SELECT ROLE,
       MIN(SALARY) AS MIN_SALARY,
       MAX(SALARY) AS MAX_SALARY
FROM emp_record_table
GROUP BY ROLE
ORDER BY MAX_SALARY DESC;
```

Output:

	ROLE	MIN_SALARY	MAX_SALARY
▶	PRESIDENT	16500	16500
	MANAGER	8500	11000
	LEAD DATA SCIENTIST	8500	9000
	SENIOR DATA SCIENTIST	5500	7700
	ASSOCIATE DATA SCIENTIST	4000	5000
	JUNIOR DATA SCIENTIST	2800	3000

Observation:

The maximum salary in the company is 16500, which belongs to the President. The other salaries follow the job hierarchy: Junior 2800 to 3000, Associate 4000 to 5000, Senior 5500 to 7700, Lead 8500 to 9000 and Manager 8500 to 11000. The only overlap is between Lead Data Scientists and Managers, where Tracy Norris (Manager, 8500) earns the same as Eric Hoffman and less than William Butler, who are both Lead Data Scientists. Apart from that each role sits in its own salary band, so the roles line up with the standard job profiles.

Q7. Ranking employees by experience

Task: Assign a rank to each employee based on their years of experience.

Query:

```
SELECT EMP_ID, FIRST_NAME, LAST_NAME, ROLE, EXP,
       RANK() OVER (ORDER BY EXP DESC) AS EXP_RANK,
       DENSE_RANK() OVER (ORDER BY EXP DESC) AS EXP_DENSE_RANK
FROM emp_record_table;
```

Output:

EMP_ID	FIRST_NAME	LAST_NAME	ROLE	EXP	EXP_RANK	EXP_DENSE_RANK
E001	Arthur	Black	PRESIDENT	20	1	1
E083	Patrick	Voltz	MANAGER	15	2	2
E103	Emily	Grove	MANAGER	14	3	3
E583	Janet	Hale	MANAGER	14	3	3
E428	Pete	Allen	MANAGER	14	3	3
E612	Tracy	Norris	MANAGER	13	6	4
E010	William	Butler	LEAD DATA SCIENTIST	12	7	5
E005	Eric	Hoffman	LEAD DATA SCIENTIST	11	8	6
E057	Dorothy	Wilson	SENIOR DATA SCIENTIST	9	9	7
E204	Karene	Nowak	SENIOR DATA SCIENTIST	8	10	8
E260	Roy	Collins	SENIOR DATA SCIENTIST	7	11	9
E245	Nian	Zhen	SENIOR DATA SCIENTIST	6	12	10
E052	Dianna	Wilson	SENIOR DATA SCIENTIST	6	12	10
E505	Chad	Wilson	ASSOCIATE DATA SCIENTIST	5	14	11
E403	Steve	Hoffman	ASSOCIATE DATA SCIENTIST	4	15	12
E478	David	Smith	ASSOCIATE DATA SCIENTIST	3	16	13
E532	Claire	Brennan	ASSOCIATE DATA SCIENTIST	3	16	13
E620	Katrina	Allen	JUNIOR DATA SCIENTIST	2	18	14
E640	Jenifer	Jhones	JUNIOR DATA SCIENTIST	1	19	15

Observation:

Arthur Black has the most experience (20 years) and is ranked 1. Three managers have 14 years each, so they share rank 3. With RANK() the next employee jumps to rank 6, while DENSE_RANK() continues at 4. I kept both columns to show the difference.

Q8. View: employees earning more than 6000

Task: Create a view that displays employees in various countries whose salary is more than six thousand.

Query:

```
CREATE VIEW emp_salary_above_6000 AS
SELECT EMP_ID, FIRST_NAME, LAST_NAME, COUNTRY, SALARY
FROM emp_record_table
WHERE SALARY > 6000;
```

Query:

```
SELECT * FROM emp_salary_above_6000
ORDER BY COUNTRY, SALARY DESC;
```

Output:

	EMP_ID	FIRST_NAME	LAST_NAME	COUNTRY	SALARY
▶	E103	Emily	Grove	CANADA	10500
	E245	Nian	Zhen	CHINA	6500
	E583	Janet	Hale	COLOMBIA	10000
	E010	William	Butler	FRANCE	9000
	E428	Pete	Allen	GERMANY	11000
	E204	Karene	Nowak	GERMANY	7500
	E612	Tracy	Norris	INDIA	8500
	E260	Roy	Collins	INDIA	7000
	E001	Arthur	Black	USA	16500
	E083	Patrick	Voltz	USA	9500
	E005	Eric	Hoffman	USA	8500
	E057	Dorothy	Wilson	USA	7700

Observation:

12 employees earn more than 6000 and they are spread across 7 countries. Since this is a view, it can be queried like a normal table and the salary filter does not have to be written again each time.

Q9. View: country wise employees earning more than 6000

Task: Create a view that lists employees across different countries who earn a salary greater than 6000.

Q8 and Q9 ask for almost the same thing, so for this one I built the view to group the same employees by country, with a count and their names in one column using GROUP_CONCAT.

Query:

```
CREATE VIEW country_wise_high_earners AS
SELECT COUNTRY,
       COUNT(EMP_ID) AS NO_OF_EMPLOYEES,
       GROUP_CONCAT(CONCAT(FIRST_NAME, ' ', LAST_NAME)
                    ORDER BY SALARY DESC SEPARATOR ', ') AS EMPLOYEES
FROM emp_record_table
WHERE SALARY > 6000
GROUP BY COUNTRY;
```

Query:

```
SELECT * FROM country_wise_high_earners
ORDER BY NO_OF_EMPLOYEES DESC;
```

Output:

	COUNTRY	NO_OF_EMPLOYEES	EMPLOYEES
▶	USA	4	Arthur Black, Patrick Voltz, Eric Hoffman, Dorothy Wilson
	GERMANY	2	Pete Allen, Karene Nowak
	INDIA	2	Tracy Norris, Roy Collins
	CANADA	1	Emily Grove
	CHINA	1	Nian Zhen
	COLOMBIA	1	Janet Hale
	FRANCE	1	William Butler

Observation:

USA has the most employees earning above 6000 (4), followed by Germany and India with 2 each. Canada, China, Colombia and France have one each.

Q10. Index on FIRST_NAME

Task: Create an index to improve the cost and performance of the query to find the employee whose FIRST_NAME is Eric, after checking the execution plan.

Execution plan before the index

Query:

```
EXPLAIN SELECT * FROM emp_record_table WHERE FIRST_NAME = 'Eric';
```

Output:

	id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
▶	1	SIMPLE	emp_record_table	NULL	ALL	NULL	NULL	NULL	NULL	19	10.0	Using where

Creating the index

Query:

```
CREATE INDEX idx_first_name ON emp_record_table (FIRST_NAME(20));
```

Execution plan after the index

Query:

```
EXPLAIN SELECT * FROM emp_record_table WHERE FIRST_NAME = 'Eric';
```

Output:

	id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
▶	1	SIMPLE	emp_record_table	NULL	ref	idx_first_name	idx_first_name	83	const	1	100.0	Using where

Query:

```
SELECT * FROM emp_record_table WHERE FIRST_NAME = 'Eric';
```

Output:

	EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT	SALARY	EMP_RATING	MANAGER_ID	PROJ_ID
▶	E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA	8500	3	E103	P105

Observation:

Before the index, the execution plan shows type ALL, which is a full table scan: MySQL went through all 19 rows to find Eric. After creating the index, the type changes to ref and idx_first_name is used as the key, with only 1 row examined and filtered at 100%. FIRST_NAME was imported as a TEXT column, so the index needs a prefix length, and 20 characters is more than enough for the names in this table. With 19 rows the speed difference is very small, but on a bigger employee table this avoids scanning the whole table.

Q11. Average salary by continent and country

Task: Calculate the average salary distribution based on the continent and country.

Query:

```
SELECT CONTINENT, COUNTRY,
       COUNT(EMP_ID) AS NO_OF_EMPLOYEES,
       ROUND(AVG(SALARY), 2) AS AVG_SALARY
FROM emp_record_table
GROUP BY CONTINENT, COUNTRY
ORDER BY CONTINENT, AVG_SALARY DESC;
```

Output:

	CONTINENT	COUNTRY	NO_OF_EMPLOYEES	AVG_SALARY
▶	ASIA	CHINA	1	6500.00
	ASIA	INDIA	3	6166.67
	EUROPE	FRANCE	1	9000.00
	EUROPE	GERMANY	3	7600.00
	NORTH AMERICA	USA	5	9440.00
	NORTH AMERICA	CANADA	3	7000.00
	SOUTH AMERICA	COLOMBIA	3	5600.00

Query:

```
SELECT CONTINENT,
       COUNT(EMP_ID) AS NO_OF_EMPLOYEES,
       ROUND(AVG(SALARY), 2) AS AVG_SALARY
FROM emp_record_table
GROUP BY CONTINENT
ORDER BY AVG_SALARY DESC;
```

Output:

	CONTINENT	NO_OF_EMPLOYEES	AVG_SALARY
▶	NORTH AMERICA	8	8525.00
	EUROPE	4	7950.00
	ASIA	4	6250.00
	SOUTH AMERICA	3	5600.00

Observation:

North America has the highest average salary (8525) and South America the lowest (5600, which is only Colombia). At country level the USA is highest at 9440, but this is pulled up by the President's salary of 16500. China and France have only one employee each, so their average is just that one salary.

Q12. Additional: bonus cost and training needs

The tasks section of the problem statement also mentions estimating bonus cost and finding employees who need training, so I added these queries as well. The bonus is taken as 5% of salary multiplied by the employee rating.

Bonus per employee (5% of salary x employee rating)

Query:

```
SELECT EMP_ID, FIRST_NAME, LAST_NAME, ROLE, SALARY, EMP_RATING,
       ROUND(0.05 * SALARY * EMP_RATING, 2) AS BONUS
FROM emp_record_table
ORDER BY BONUS DESC;
```

Output:

EMP_ID	FIRST_NAME	LAST_NAME	ROLE	SALARY	EMP_RATING	BONUS
E001	Arthur	Black	PRESIDENT	16500	5	4125.00
E083	Patrick	Voltz	MANAGER	9500	5	2375.00
E428	Pete	Allen	MANAGER	11000	4	2200.00
E103	Emily	Grove	MANAGER	10500	4	2100.00
E204	Karene	Nowak	SENIOR DATA SCIENTIST	7500	5	1875.00
E612	Tracy	Norris	MANAGER	8500	4	1700.00
E052	Dianna	Wilson	SENIOR DATA SCIENTIST	5500	5	1375.00
E005	Eric	Hoffman	LEAD DATA SCIENTIST	8500	3	1275.00
E260	Roy	Collins	SENIOR DATA SCIENTIST	7000	3	1050.00
E583	Janet	Hale	MANAGER	10000	2	1000.00
E010	William	Butler	LEAD DATA SCIENTIST	9000	2	900.00
E478	David	Smith	ASSOCIATE DATA SCIENTIST	4000	4	800.00
E403	Steve	Hoffman	ASSOCIATE DATA SCIENTIST	5000	3	750.00
E245	Nian	Zhen	SENIOR DATA SCIENTIST	6500	2	650.00
E640	Jenifer	Jhones	JUNIOR DATA SCIENTIST	2800	4	560.00
E505	Chad	Wilson	ASSOCIATE DATA SCIENTIST	5000	2	500.00
E057	Dorothy	Wilson	SENIOR DATA SCIENTIST	7700	1	385.00
E532	Claire	Brennan	ASSOCIATE DATA SCIENTIST	4300	1	215.00
E620	Katrina	Allen	JUNIOR DATA SCIENTIST	3000	1	150.00

Query:

```
SELECT ROUND(SUM(0.05 * SALARY * EMP_RATING), 2) AS TOTAL_BONUS_COST
FROM emp_record_table;
```

Output:

TOTAL_BONUS_COST
23985.00

Employees flagged for training (rating below 2)

Query:

```
SELECT EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPT, EMP_RATING
FROM emp_record_table
WHERE EMP_RATING < 2;
```

Output:

	EMP_ID	FIRST_NAME	LAST_NAME	ROLE	DEPT	EMP_RATING
▶	E057	Dorothy	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	1
	E532	Claire	Brennan	ASSOCIATE DATA SCIENTIST	AUTOMOTIVE	1
	E620	Katrina	Allen	JUNIOR DATA SCIENTIST	RETAIL	1

Observation:

The total bonus cost comes to 23985. Arthur Black gets the highest bonus (4125). Three employees are rated 1 and should be the first ones considered for training: Dorothy Wilson, Claire Brennan and Katrina Allen. Dorothy is a Senior Data Scientist with 9 years of experience, so her low rating is worth looking into with her manager.

Summary

The company has 19 employees across four departments, and most of them are rated between 2 and 4. Salaries match the role hierarchy, with only a small overlap between Lead Data Scientists and Managers. The highest salary is 16500 (President). Three employees with a rating of 1 are candidates for training, and the estimated bonus cost at 5% of salary times rating is 23985.